

Cinema Log & Release Radar — Use Case Model

CP353002 · Section 03 · source: doc/diagrams/01-use-case.puml

1. Use Case Diagram

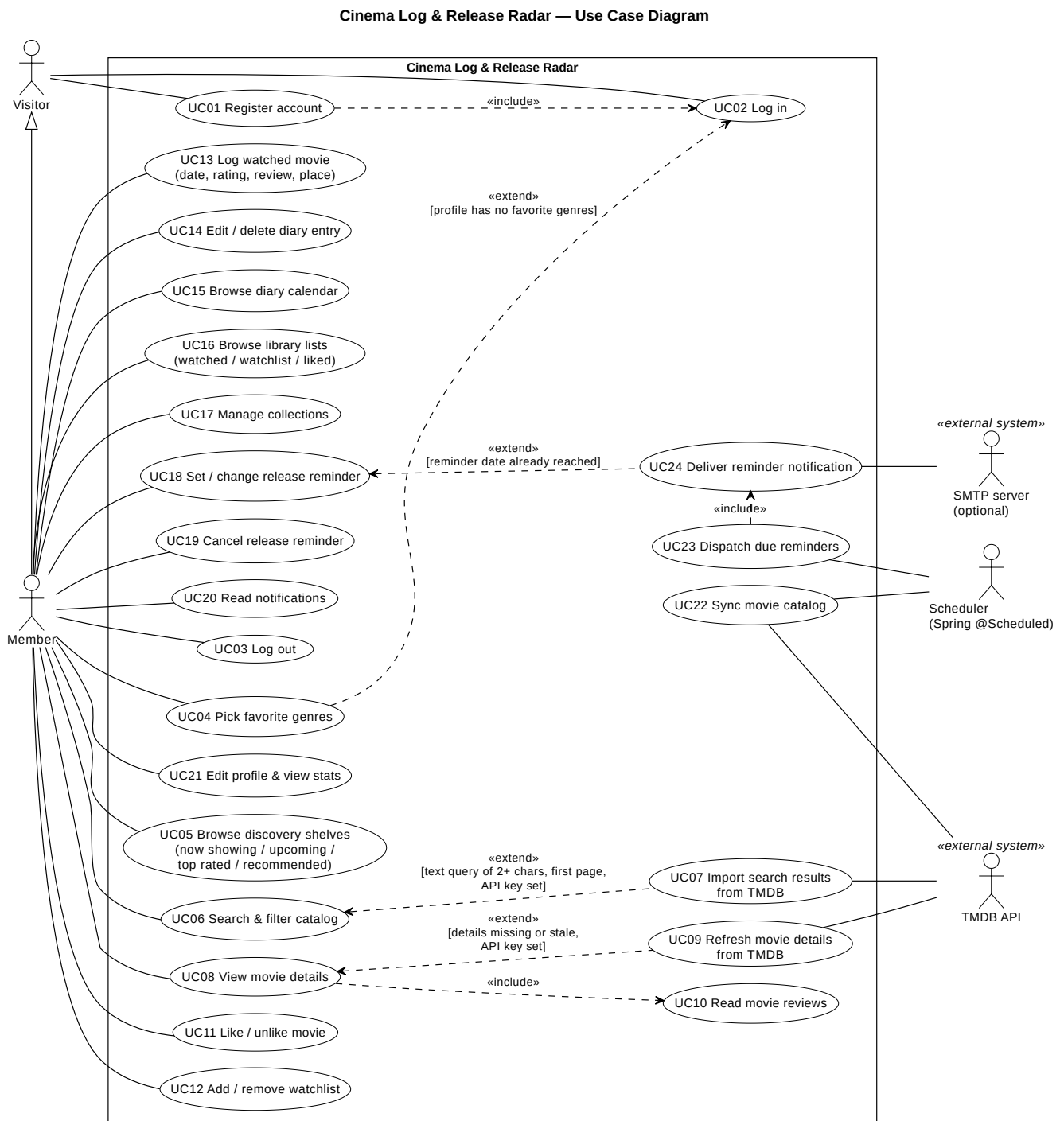


Figure 1 — Use case diagram

2. Actors

Actor	Description
Visitor	Not logged in. Can only reach /login, /register and the public API (SecurityConfig).
Member	A logged-in Visitor (generalization). Every page and /api/v1/users/me/** needs a session.
Scheduler	Spring @Scheduled jobs MovieSyncJob and ReminderJob. Each also runs once on ApplicationReadyEvent.
TMDB API, SMTP server	External systems. Email is optional (spring.mail.host may be empty).

3. «include» / «extend» relationships

Relation	Why	Code
UC01 «include» UC02	After a successful register the page always submits the hidden login form	templates/auth/register.html (#auto-login), pages/auth.js
UC04 «extend» UC02	Only when the profile has no favorite genres	OnboardingAwareSuccessHandler → UserProfileService.needsOnboarding()
UC07 «extend» UC06	Only for a text query of 2+ characters, on page 0, with a TMDB key	MovieQueryServiceImpl.search() → MovieSyncService.importSearchResults()
UC08 «include» UC10	The movie page always loads its reviews	pages/movie.js → GET /api/v1/movies/{id}/reviews
UC09 «extend» UC08	Only when details are missing or stale and a key is set	MovieQueryServiceImpl.getDetail() → Movie.needsDetails() → refreshDetails()
UC23 «include» UC24	Every due reminder is delivered	ReminderDispatchServiceImpl.dispatchDueReminders()
UC24 «extend» UC18	Only when the reminder date has already been reached	ReminderServiceImpl.save() → dispatchIfDue()

4. Use Case Descriptions

Use Case ID	UC01
Use Case Name	Register account
Primary Actor	Visitor
Description	A visitor creates an account and is logged in straight away.
Pre-conditions	Not logged in
Post-conditions	users + user_profiles rows exist; password stored as BCrypt hash
Main Flow	- Visitor enters display name, email and password. - Page sends POST /api/v1/auth/register. - System creates the account and profile. - Page logs the visitor in automatically (UC02).
Alternative Flow	- Email already used → 409. - Blank name, invalid email, or password not 6–72 characters → 400 with fieldErrors.

Use Case ID	UC02
Use Case Name	Log in
Primary Actor	Visitor
Description	A visitor signs in with email and password and gets a session.
Pre-conditions	Account exists
Post-conditions	Session cookie set
Main Flow	- Visitor submits email + password to /login. - System creates a session. - System redirects to /films.
Alternative Flow	- Wrong email/password → /login?error. - No favorite genres → /onboarding (UC04).

Use Case ID	UC03
Use Case Name	Log out
Primary Actor	Member
Description	A member ends the current session.
Pre-conditions	Logged in
Post-conditions	Session ended, redirect to /login?logout
Main Flow	Member opens the nav menu and chooses log out (POST /logout).
Alternative Flow	—

Use Case ID	UC04
Use Case Name	Pick favorite genres
Primary Actor	Member
Description	A member chooses the genres used for recommendations.
Pre-conditions	• Logged in
Post-conditions	• user_favorite_genres replaced
Main Flow	1. Member ticks genres. 2. Page sends PUT /api/v1/users/me/favorite-genres.
Alternative Flow	• None ticked → 400 "Pick at least one genre". • More than 10 → 400.

Use Case ID	UC05
Use Case Name	Browse discovery shelves
Primary Actor	Member
Description	A member browses the home shelves of movies.
Pre-conditions	• Logged in
Post-conditions	—
Main Flow	1. Member opens /films. 2. System shows now showing, upcoming, top rated and recommended (/api/v1/movies/now-showing, /upcoming, /top-rated, /recommended).
Alternative Flow	• No favorite genres → recommended falls back to top rated.

Use Case ID	UC06
Use Case Name	Search & filter catalog
Primary Actor	Member
Description	A member searches the catalog with text and filters.
Pre-conditions	• Logged in
Post-conditions	—
Main Flow	1. Member sets text, genre, year, decade, released-before, min rating, language and sort. 2. Page sends GET /api/v1/movies. 3. System returns 40 results per page.
Alternative Flow	• No match → empty state. • TMDB error → local results only (warning logged).

Use Case ID	UC07
Use Case Name	Import search results from TMDB
Primary Actor	System (extends UC06)
Description	During a search the system imports matching movies from TMDB.
Pre-conditions	Text query of 2+ characters (<code>MovieSearchCriteria.hasText()</code>), page 0, TMDB key set
Post-conditions	New/updated movies rows
Main Flow	<code>TmdbMovieCatalogAdapter.search()</code> is called. - System upserts matches by <code>tmdb_id</code>.
Alternative Flow	TMDB unreachable → <code>ExternalServiceException</code> caught, search continues.

Use Case ID	UC08
Use Case Name	View movie details
Primary Actor	Member
Description	A member opens a movie page with details and actions.
Pre-conditions	Logged in
Post-conditions	—
Main Flow	- Member opens <code>/movies/{id}</code>. - System shows details, cast, reviews (UC10) and similar movies. - Page shows buttons for like, watchlist, watched, rate, review, remind and collection.
Alternative Flow	- Unknown id → 404. - Movie not released yet → watched/rate/review disabled, remind enabled.

Use Case ID	UC09
Use Case Name	Refresh movie details from TMDB
Primary Actor	System (extends UC08)
Description	When details are missing or stale the system refreshes them from TMDB.
Pre-conditions	Details missing or stale, TMDB key set
Post-conditions	<code>details_fetched_at</code> updated
Main Flow	<code>MovieSyncService.refreshDetails()</code> is called. - Runtime, director, cast and genres are updated.
Alternative Flow	TMDB error → old data shown.

Use Case ID	UC10
Use Case Name	Read movie reviews
Primary Actor	Member
Description	A member reads the reviews of a movie.
Pre-conditions	• Movie exists
Post-conditions	—
Main Flow	1. Page sends GET /api/v1/movies/{id}/reviews. 2. System returns the average rating and reviews, own review first.
Alternative Flow	• Unknown movie → 404.

Use Case ID	UC11
Use Case Name	Like / unlike movie
Primary Actor	Member
Description	A member likes or unlikes a movie.
Pre-conditions	• Logged in
Post-conditions	• liked_movies row added/removed
Main Flow	1. Member presses ♡ → PUT /api/v1/users/me/likes/{movieId}; pressing it again → DELETE.
Alternative Flow	• Already liked → no change. • Unknown movie → 404.

Use Case ID	UC12
Use Case Name	Add / remove watchlist
Primary Actor	Member
Description	A member adds a movie to or removes it from the watchlist.
Pre-conditions	• Logged in
Post-conditions	• watchlist_items row added/removed
Main Flow	1. Member presses + → PUT /api/v1/users/me/watchlist/{movieId}; remove → DELETE.
Alternative Flow	• Already on list → no change. • Unknown movie → 404.

Use Case ID	UC13
Use Case Name	Log watched movie
Primary Actor	Member
Description	A member records that they watched a movie, with optional rating and review.
Pre-conditions	Logged in, movie released
Post-conditions	watched_movies row added; movie removed from watchlist; if a review was written, an in-app "review added" notification
Main Flow	- Member picks date, stars, review and place. - Page sends POST /api/v1/users/me/diary.
Alternative Flow	- Date in the future or movie not out yet → 400. - Rating outside 1–5 → 400. - Unknown movie → 404.

Use Case ID	UC14
Use Case Name	Edit / delete diary entry
Primary Actor	Member
Description	A member changes or deletes one of their diary entries.
Pre-conditions	Entry belongs to the member
Post-conditions	Entry updated (first review added → notification) or removed
Main Flow	Member chooses edit → PUT /api/v1/users/me/diary/{id}, or delete → DELETE /api/v1/users/me/diary/{id}.
Alternative Flow	- Same date rules as UC13. - Entry of another user → 404.

Use Case ID	UC15
Use Case Name	Browse diary calendar
Primary Actor	Member
Description	A member browses their diary as a month calendar.
Pre-conditions	Logged in
Post-conditions	—
Main Flow	- Member opens /diary month grid (/diary/calendar?month=). - Member clicks a day (/diary/day?date=).
Alternative Flow	Empty month → empty state.

Use Case ID	UC16
Use Case Name	Browse library lists
Primary Actor	Member
Description	A member views their watched, watchlist and liked lists.
Pre-conditions	• Logged in
Post-conditions	—
Main Flow	1. Member opens the /watched, /watchlist or /liked page.
Alternative Flow	• Empty list → empty state.

Use Case ID	UC17
Use Case Name	Manage collections
Primary Actor	Member
Description	A member creates, renames and deletes collections and adds or removes movies in them.
Pre-conditions	• Logged in
Post-conditions	• collections / collection_movies changed
Main Flow	1. Create, rename, delete via /api/v1/users/me/collections. 2. Add/remove movie via PUT/DELETE /{id}/movies/{movieId}.
Alternative Flow	• Duplicate name (case-insensitive) → 409. • Blank name → 400. • Removing a movie not in it → 404.

Use Case ID	UC18
Use Case Name	Set / change release reminder
Primary Actor	Member
Description	A member asks to be reminded before an upcoming movie is released.
Pre-conditions	• Movie not released yet
Post-conditions	• reminders row SCHEDULED (or SENT)
Main Flow	1. Member chooses 7 / 3 / 1 / 0 days and IN_APP or EMAIL. 2. Page sends PUT /api/v1/users/me/reminders/{movieId}.
Alternative Flow	• Other offset → 400. • Unknown movie → 404. • Movie already out or no release date → 400. • Reminder already exists → rescheduled. • Date already reached → delivered now (UC24).

Use Case ID	UC19
Use Case Name	Cancel release reminder
Primary Actor	Member
Description	A member removes a reminder.
Pre-conditions	Reminder exists
Post-conditions	Status CANCELLED
Main Flow	1. Page sends DELETE /api/v1/users/me/reminders/{movieId}.
Alternative Flow	No reminder → 404.

Use Case ID	UC20
Use Case Name	Read notifications
Primary Actor	Member
Description	A member reads their notifications from the bell menu.
Pre-conditions	Logged in
Post-conditions	All notifications is_read = true
Main Flow	1. Bell shows unread count + latest 10. 2. Opening the bell while there are unread items → POST /notifications/read-all.
Alternative Flow	—

Use Case ID	UC21
Use Case Name	Edit profile & view stats
Primary Actor	Member
Description	A member edits their profile and sees their statistics.
Pre-conditions	Logged in
Post-conditions	users.email / user_profiles updated
Main Flow	1. Member edits name, email, bio, avatar and favorite genres → PUT /api/v1/users/me. 2. Profile page shows /api/v1/users/me/stats.
Alternative Flow	- Email used by another account → 409. - Invalid fields → 400.

Use Case ID	UC22
Use Case Name	Sync movie catalog
Primary Actor	Scheduler
Description	A scheduled job keeps the local catalog in step with TMDB.
Pre-conditions	• TMDB key set
Post-conditions	• Missing genres added (existing names kept); movies and movie_genres upserted
Main Flow	1. MovieSyncJob runs syncGenres() + syncCatalog(). 2. Now playing, upcoming, popular, top rated × sync-pages are fetched. 3. Movies are upserted by tmdb_id.
Alternative Flow	• No key → skipped silently. • TMDB error → WARN "TMDB sync skipped".

Use Case ID	UC23
Use Case Name	Dispatch due reminders
Primary Actor	Scheduler
Description	A scheduled job sends every reminder whose date has been reached.
Pre-conditions	—
Post-conditions	• Reminders SENT
Main Flow	1. ReminderJob calls dispatchDueReminders(). 2. SCHEDULED reminders with date ≤ today are found. 3. UC24 runs for each one.
Alternative Flow	• Job error → logged.

Use Case ID	UC24
Use Case Name	Deliver reminder notification
Primary Actor	System
Description	The system delivers one due reminder through its channel.
Pre-conditions	• Reminder due
Post-conditions	• notifications row added; reminder SENT
Main Flow	1. Publish ReminderDueEvent. 2. NotificationSenderFactory.forChannel() picks the sender. 3. Sender send() delivers it. 4. A notifications row is saved.
Alternative Flow	• EMAIL without mail server → email skipped (logged), record still saved.